



Doping and Ergogenic Aids

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SPORTS PHARMACOLOGY: ERGOGENIC DRUGS IN SPORT

There is no room for second place. There is only one place in my game, and that's first place.

Vince Lombardi¹

In this day and age, it is common for athletes to be scrutinized for using prohibited performance-enhancing drugs and methods. Our society places a high value on winning and rewards athletes who perform well. As a result, competitors are often motivated to do “whatever it takes” to perform at a higher level and thus improve in their sport. The word *ergogenic* is derived from the Greek *éaargon*, “to work,” and *gennan*, “to produce”; it literally means “increasing the ability to do work.” Athletes have sought ways to improve their performance for thousands of years. Athletes at the first Greek Olympics ate substances that they thought would give them a competitive advantage, including dried figs, mushrooms, large amounts of meat, and strychnine (a known poison). These ancient Greek athletes were willing to risk taking a potentially deadly substance in the hope of winning. History does repeat itself, because this trend continues today and has escalated to incomprehensible levels. The ensuing battle to weed out athletes who engage in unfair practices is a challenge to organizations that believe in the concepts of “an even playing field” and fairness in sport.

The desire to win at all costs was confirmed by a poll of Olympic athletes conducted by Mirkin² in the 1980s. The question was, “If you could take a pill that would guarantee you the Olympic gold medal but would kill you within a year, would you take it?” The shocking result was that more than 50% of the athletes would take the pill.

Goldman³ subsequently conducted a modified poll in the 1990s and asked 198 aspiring Olympians, all elite athletes, two questions:

1. If you were offered a banned performance-enhancing substance that guaranteed that you would win an Olympic medal and you could not be caught, would you take it?
2. If you were guaranteed that you would not be caught, would you take a banned performance-enhancing drug that would allow you to win every competition for the next 5 years but would then cause you to die from the side effects of the substance?

Of 198 athletes, 195 answered yes to the first question; in response to the second question, more than 50% of the athletes said they would take the substance.

The results of these two polls reveal that a high percentage of athletes are willing to use substances that are potentially fatal if they believe that those substances will help them win. Athletes want to win *now* and may not consider or even be concerned about the future consequences of taking a performance-enhancing substance.

Testosterone: Historical Perspectives

In the 1950s it was discovered that Olympic weight lifters were gaining immense strength by injecting testosterone, a male hormone. An American physician, John Bosley Ziegler, saw the potential benefits, and in an effort to minimize the adverse androgenic effects that the weight lifters were experiencing, he developed the derivative methandrostenolone.⁴ Although it was initially considered a superior ergogenic drug compared with testosterone, it had its own adverse effects related to its potent androgenic properties. Since then many scientists have attempted and failed to develop a pure anabolic agent without androgenic features. To underscore the important point that derivatives of testosterone have anabolic and androgenic properties, we categorize all ergogenic derivatives of testosterone in this chapter as anabolic-androgenic steroids (AASs). Hundreds of ergogenic variants of testosterone are now being used by athletes.

One of the first scientists to promote the ergogenic properties of testosterone was the prominent French physiologist Brown-Séquard.⁵ Although largely remembered in the medical world today for his description of a spinal cord syndrome, Brown-Séquard played an important role in the history of AASs. In June 1889, at the Société de Biologie meeting in Paris, the 72-year-old Brown-Séquard announced that he had found a “rejuvenating compound” that had reversed many of his physical and mental ailments related to aging. He reported that he had injected himself with a liquid extract made from the testicles of dogs and guinea pigs and that these injections had dramatically increased his strength, improved his mental acumen, relieved his constipation, and increased the arc of his urinary stream. Brown-Séquard’s findings were viewed with disdain, yet he had made an important discovery with enormous implications for the medical and athletic worlds. Brown-Séquard’s discovery of the positive properties of the testicular extract was based on the earlier work of Berthold, who had proposed that the implantation

Abstract

The plethora of available substances touting improved performance—be it powders, supplements, even vitamins—is widespread and growing. However, these aids are not without dangers and oftentimes can result in illness, worsened execution of the activities, disqualification from sport, and even death. Some of these are well known, such as testosterone and growth hormone, due to their role in publicly visible abuses across the long history of sports. Others, such as caffeine and alcohol, are commonly used and thus their effects on athletics is overlooked or not considered. Individuals interested in exploring and utilizing performance enhancing products should do so with a clear understanding of the benefits as well as the potential hazards.

Keywords

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of testicles in the abdomen of roosters reversed the effects of castration.⁶

Scientists continued the work of these two scientists, and several important discoveries were made that contributed to a better understanding of the properties of the testicular hormone. The Dutch scientist David and his colleagues first isolated testosterone in 1935.⁷ Soon thereafter, German scientists published an article describing the synthesis of testosterone from cholesterol.⁸ Subsequently, physicians and veterinarians started using testosterone and its intermediaries clinically,⁹ and testosterone was first used medically in humans to help chronically ill patients recover muscle mass.¹⁰ Testosterone was found to have positive effects on almost every organ of the body, and 60 years after Brown-Séquard announced his discovery, testosterone was promoted once again as a rejuvenation compound. In 1939 Boje¹⁰ suggested that male sex hormones might improve athletic performance. In 1941, after a racehorse dramatically improved its performance following injections of testosterone,¹¹ human competitors who were looking for an advantage during competition started experimenting with testosterone use.¹²

The first systematic use of testosterone in sports was by Soviet Olympic athletes in the 1950s; by the 1960s, use of AASs was increasingly prevalent.¹³ During that time, no sanctions against the use of ergogenic drugs existed, and many athletes believed that it was necessary to use drugs to be able to remain competitive. By the 1968 Olympics, the use of AASs was pervasive across various disciplines of sport, including both strength and aerobic events. New drugs claiming to have superior anabolic qualities were being marketed openly, and male athletes in large numbers from other sports such as American football soon began taking AASs either knowingly or allegedly as “vitamins.”¹⁴ It was estimated that 40% to 90% of professional football players were using AASs by the 1980s,¹⁵ and soon thereafter, college football players followed suit.¹⁵

Female athletes began experimenting with the drugs to improve their own performance abilities. Before long, it was discovered that women who took AASs experienced increased muscular strength. In addition, the androgenic properties of AASs also induced masculinizing effects, including deepening of the voice, amplification of body hair, loss of breast tissue, and clitoral enlargement. The masculinization of female athletes from the Soviet Bloc nations was so dramatic that gender tests and chromosomal analyses were performed at the 1967 European Championships to ensure that male athletes were not masquerading as female athletes.

Women in Western nations have been suspected of using ergogenic drugs. Florence Griffith Joyner became one of the most famous athletes in the world when she reappeared on the world stage after coming out of semiretirement and achieved tremendous success at the 1988 Olympic Games. Joyner had progressed from a good runner to a world champion whose record in the 100-m run was unsurpassed for 10 years. Her muscular physique, combined with her newfound athletic success, led to speculation that her athletic superiority was a result of the use of performance-enhancing drugs—a charge she vehemently denied.¹⁶ Joyner instead insisted that her athletic prowess was the result of a relentless training program supervised by her

husband, Olympian Al Joyner. When Florence Griffith Joyner died in her sleep at the young age of 38 years, speculation that her death may have been related to adverse effects from taking ergogenic drugs was rampant. To date, it has not been verified that Joyner used ergogenic aids or that her death was related to the use of drugs, and her death has been attributed to a seizure.

Many young athletes believe that the use of ergogenic drugs is a worthwhile risk. An eye-opening study by Buckley and colleagues¹⁷ revealed that 6.6% of high school seniors had, at some point, used anabolic steroids. Of these, 30% of users were non-athletes using ergogenic drugs to improve their appearance rather than their athletic performance. Evidence indicates that ergogenic drug abuse commences in middle school. The 2013 survey by the Partnership for Drug-Free Kids revealed that 11% of high school teens had tried synthetic human growth hormone (GH) at least once without a prescription—an increase of 5% in just 1 year!¹⁸ Furthermore, Yesalis and Bahrke¹⁹ found that the percentage of 14- to 18-year-old *girls* using anabolic steroids had almost doubled in 7 years, as many considered the use an effective means of obtaining college athletic scholarships.²⁰

When a Danish cyclist died at the 1960 Olympics after ingesting a combination of nicotinic acid and amphetamines, concerns were raised regarding both fairness and safety, and a plan of action to address these issues was formulated. One of the first steps was to define the problem, and in 1963, the Council of Europe established a definition of doping as “the administration or use of substances in any form alien to the body or of physiologic substances in abnormal amounts and with abnormal methods by healthy persons with the exclusive aim of attaining an artificial and unfair increase in performance in competition.”¹⁸

Anabolic-Androgenic Steroids

Physiologic Considerations

The three main steroids in the human body are androgens, estrogens, and corticosteroids. The androgens are responsible for the development of male characteristics, whereas the estrogens express female characteristics. Corticosteroids—both glucocorticoids and mineralocorticoids—are responsible for a wide variety of essential body functions involving the immune, cardiovascular, metabolic, and hemostatic systems.

Females produce small amounts of androgens in the ovary and adrenal gland, and males produce small amounts of estrogen, but it is the production of androgens that makes males “masculine” and the production of estrogen that makes females “feminine.” The most abundant androgen in males is testosterone, which is produced primarily in the testes. In many target cells of the body, testosterone is reduced at the 5 α position to dihydrotestosterone, which serves as the intracellular mediator of the actions of testosterone.²¹ Dihydrotestosterone exhibits a greater affinity for androgen receptors compared with testosterone and is thus the more stable and potent androgen. Precursors in the metabolic pathway to testosterone, dehydroepiandrosterone (DHEA) and androstenedione, bind less strongly to the androgen receptors and are known as weak androgens. Other weak androgens that are metabolites of testosterone include etiocholanolone and androsterone. Although they are weak, these androgens can induce significant anabolic effects.

In males, testosterone production peaks at three distinct phases of life. The first peak occurs during the fetal period, in the second trimester, and directs male fetal development. Subsequently, a smaller surge occurs during the first year of life and the largest surge occurs during puberty, resulting in several visible changes. Androgens cause laryngeal enlargement, thus deepening the voice; genital maturity; spermatogenesis in the testicles; and bone and muscle growth, with a relative decrease in body fat. The skin becomes thicker and oilier, and body hair on the face and in the axillae and groin grows to adult levels.

On the cellular level, testosterone, or its more active metabolite, dihydrotestosterone, works by binding to an intracellular androgen receptor in the cytoplasm. The steroid-androgen receptor complex is transported to the nucleus, where it attaches to a specific hormone regulatory complex on nuclear chromosomes. Here the complex stimulates the synthesis of specific ribonucleic acids and proteins. The ribonucleic acid compounds are transported through the bloodstream to act on target organs and stimulate spermatogenesis, effect sexual differentiation, and increase protein synthesis in muscle tissues. The androgenic message is received and processed in all organs that have androgenic receptors.

The metabolism of testosterone and its products occurs rapidly in the liver. Oral testosterone is absorbed immediately and metabolized so rapidly that it becomes ineffective. Scientists have therefore developed methods of altering the basic structure of the molecule to delay its metabolism and increase its half-life in the plasma. Alkylation at the 17 α position with a methyl or ethyl group allows oral agents to be degraded slowly. Similarly, esterification of the 17 β position allows parenteral agents to resist degradation. Testosterone esters, such as cypionate and enanthate esters, are more potent than testosterone. These compounds must be injected intramuscularly, usually at 1- to 3-week intervals. Newer preparations of testosterone can be administered transdermally.²² These preparations were originally applied to the scrotum, but new transdermal patches and creams can be applied to other parts of the body.

In addition to promoting muscle development, androgens cause the skin to become unusually oily, which can lead to acne, increase body hair, and induce male pattern baldness, with hair-line recession and thinning and loss of central scalp hair. The skin effects are crucial clues for physicians who suspect anabolic steroid use. The extent of the acne can be profound, and the back is often significantly affected. Lastly, the distinctive odor caused by the effects of anabolic steroids on the sebaceous glands is inimitable.

Athletic Performance Considerations

Despite the perceptions of athletes, scientific data regarding the effects of anabolic steroids on athletic performance are varied. When used at therapeutic doses, neither strength nor performance gains are expected, because in a homeostatic state, hormonal levels are maintained within a narrow window. After a therapeutic dose of a hormone is administered, the body halts its own endogenous hormone production to maintain constant levels. Athletes who are cognizant of this fact have taken doses 50 to 100 times the therapeutic dose, which shuts down endogenous production

of AASs. But does taking such high doses improve athletic performance?

In the world of athletics, several popular theories foster steroid abuse. The first theory is that "In combination with training, AASs increase lean body mass and decrease body fat." Early studies that evaluated this theory showed equivocal results.²³ More recent studies have shown an increase in body weight and lean body mass but no significant decrease in the percentage of body fat.²⁴

Another theory suggests that "AASs increase muscle strength." Early studies in which mostly low-dose AASs were evaluated were inconclusive.²⁵ Later, in 1996, Bhasin and colleagues²⁶ studied the effect of supraphysiologic doses of testosterone on healthy men; this was the first prospective randomized study that looked at the effect of "super dosing" of AASs. In this study of 43 healthy men, the group that received steroids had definite increases in strength and muscle size compared with the placebo group. Notably, no significant adverse effects were observed in the steroid group; however, the study had limitations, including a short term (6 weeks) and inadequate follow-up. However, this study was the first to provide scientific evidence for the theory that athletes believed for decades—that steroids do work, especially at the amplified doses that are popular in athletics. Other investigations have shown distinct advantages for muscle, with both expansion in cross-sectional diameter and proliferation of new fibers. The upper regions of the body are more susceptible to gains from AASs because of the relatively larger number of androgen receptors in these areas. AASs, however, do not appear to improve athletic endurance.²⁴

AASs work on several different levels in the body. They stimulate not only protein synthesis but also the production of GH, a potent ergogenic agent. Furthermore, AASs display some anticatabolic features because they delay the effects of cortisol, a stress hormone. Cortisol is released in response to physical and psychological stress and causes protein degradation and muscle atrophy.²⁷ Testosterone slows this catabolism by displacing the corticosteroids from receptor sites.²⁸ Additionally, AASs may increase oxygen uptake, cardiac output, and stroke volume.²⁹ They are thought to improve athletic performance by increasing aggressive behaviors; however, the data on the relationship between testosterone levels and aggressive behavior remain inconclusive.^{30,31}

Adverse Effects

The use of anabolic steroids may result in several less than desirable side effects. Oily skin, acne, small testicles, gynecomastia, and changes in hair patterns are the most common. Small doses of androgens increase sebaceous gland secretions, leading to the changes in the skin and the development of acne.³² Enlargement of the male breast (gynecomastia) is characterized by the presence of firm glandular tissue and is usually associated with an increased production of estrogens or decreased levels of androgens. The specific pathophysiology of gynecomastia in men taking AASs remains unclear. One theory suggests that the body homeostatically shuts down the production of endogenous androgens. Once the user stops using AASs, an increase occurs in the relative level of estrogens, leading to the development of breast tissue. Gynecomastia is irreversible once levels of androgens return to normal. A means of offsetting the estrogen-related adverse effects

of AASs is to take an antiestrogenic drug such as clomiphene citrate (Clomid). Clomid blocks the effects of increased estrogen in persons using AASs by binding and thus blocking estrogen receptors.

Small testicles often also develop in men who use AASs. The testicles atrophy and shrink, which is a response to the homeostatic effect triggered by high exogenous AASs. Often the testicles remain smaller after discontinuation of the AASs. Other adverse effects on the male reproductive system include oligospermia (a decreased number of sperm) and azoospermia (the absence of sperm).³³ Moreover, taking exogenous AASs decreases not only levels of endogenous testosterone but also the circulation of follicle-stimulating hormone and luteinizing hormone, which can lead to male infertility. Testosterone-dependent prostate enlargement and accelerated growth of prostate cancer have been documented.³⁴ Women who take AASs experience virilization effects, including deepening of the voice, an increased amount of body hair, loss of breast tissue, and enlargement of the clitoris. Irregularities in the menstrual cycle as well as infertility or early menopause may result, and many of these adverse effects in women are irreversible.³⁵

Musculoskeletal. The musculoskeletal system may be affected adversely, especially in prepubertal athletes who take AASs. Although bone growth is stimulated immediately after the use of AASs, the growth plates of long bones may close prematurely. The resultant short stature is permanent.³⁶ Disproportionate changes in strength of the muscle-tendon unit (greater) than tendon strength (lesser) increases the susceptibility to tendon rupture.³⁷

Cardiovascular. Numerous adverse changes occur in the heart of a person who takes AASs. Androgens are modulators of serum lipoprotein, thus increasing plasma levels of low-density lipoproteins (“bad” lipids) and decreasing levels of high-density lipoproteins (“good” lipids). Oral and synthetic AASs have more pronounced negative effects on lipid metabolism than do injectable or natural AASs.³⁸ An altered lipid pattern may lead to atherosclerotic heart disease. Male and female users of AASs show similar adverse changes in lipid metabolism.³⁹

Animal studies have suggested that AASs may cause myocardial damage.⁴⁰ A study that investigated the relationship among resistance training, anabolic steroid use, and left ventricular function in elite bodybuilders showed concentric left ventricular hypertrophy. However, no effect on cardiac function was observed, which suggests that the heart may enlarge as a physiologic adaptation to the intensive training potentiated by use of AASs.⁴¹ In a set of bodybuilding twins, one twin used AASs for more than 15 years whereas the other abstained. No significant difference in cardiac function was found between the twins.⁴² However, noteworthy changes in cardiac function after the use of AASs have been reported in other cases, such as myocardial infarction and death in young athletes.⁴³ These case reports are no doubt worrisome yet do not provide scientific “proof” that the use of AASs directly leads to myocardial infarction or death. Nevertheless, cardiac pathology was evident upon postmortem examination in more than one-third of 34 persons (aged 20 to 45 years) who had used AASs and died as a result of accidents, suicides, and homicides.⁴⁴ Cardiac abnormalities may not resolve after

the use of AASs is stopped. A study of former users of AASs showed that many years after discontinuing their use, strength athletes showed persistent left ventricular hypertrophy compared with strength athletes who had not used AASs.⁴⁵

Hepatic. Most AAS metabolism occurs in the liver, which is thus prone to liver damage. Athletes with preexisting liver dysfunction are at greatest risk for androgen-induced damage. Temporary liver disturbances are common in athletes who use oral androgens, although function appears to return to normal after discontinuation of the drugs.^{46,47} Several investigators believe that AAS-induced hepatotoxicity is overstated, attributing resultant transaminitis to skeletal rather than liver damage.⁴⁸ A study of the effect of AASs on rats showed definite cellular damage to hepatocytes,⁴⁹ yet long-term studies on the effect of AASs on the liver in human subjects have not been conducted; until we have scientific data, hepatotoxicity should remain a potential concern.

Hepatotoxicity may manifest as peliosis hepatis, that is, the formation of blood-filled cysts in the liver. Enlarged liver cells block venous and lymphatic flow, producing cholestasis, necrosis, and peliosis cysts, which can rupture and may be life threatening. In contrast to most adverse effects of AASs, peliosis hepatitis does not appear to be dose related and can occur at any time after the use of AASs is started.⁵⁰ Pathogenesis is speculated to be from AAS-mediated hepatocyte hyperplasia.⁵¹

Last, although there is no definite evidence to support a causal relationship between hepatocellular carcinoma and steroid abuse, there have been multiple reports of the development of hepatocellular carcinoma after AAS use.⁵²

Psychiatric. The most commonly associated psychiatric effect of AASs is an increase in aggressive behaviors, although only a few well-conducted prospective randomized studies confirm this observation. In fact, most studies using therapeutic doses of exogenous testosterone showed no adverse effects, and a short-term study of supratherapeutic dosing of AASs in athletes revealed no significant psychiatric effects.⁵³ However, some studies have reported positive effects on mood.⁵⁴ In Brown-Séquard’s 19th-century study of the self-administration of testicular substrates, one of the reported positive effects was improvement in mood, which was accompanied by an enhancement of his physical stature.⁵ Subsequently, benefits of AASs on mood and other psychiatric maladies were further evaluated by physicians following the advent of synthetic testosterone derivatives in the 1930s.^{55,56} AASs were being researched during their use to treat a variety of conditions ranging from depression to psychoses, and although results were varied, some small studies revealed an increase in aggression, thought to be linked to other personality disorders such as antisocial, borderline, and histrionic personalities.⁵⁷

Additional research suggests an association with AAS use and psychiatric disturbances. A study of 41 athletes using supratherapeutic doses of AASs for an average of 45 weeks revealed that 34% experienced symptoms of major mood disorders such as severe depression or mania. AAS users are also more likely to abuse alcohol, tobacco, and illicit drugs,⁵⁸ and data suggest that AAS users tend to meet criteria for substance dependence disorder more commonly than nonusers.⁵⁹

Disorders of body image have also been reported after use of AASs. Many weight lifters who may appear muscular compared

with the average athlete consider themselves small or weak. This dysmorphic disorder is similar to that of women with anorexia nervosa who, despite being much thinner than the average woman, erroneously perceive themselves as fat.

Steroid Supplements

In 1994, the US Congress passed legislation affecting all persons involved in the care of athletes—the Dietary Supplement Health and Education Act. This law permits numerous substances to be sold without prior approval from the US Food and Drug Administration (FDA) as long as they are sold as dietary supplements and not as “drugs.”

The Dietary Supplement Health and Education Act also established a formal definition of a dietary supplement using several criteria. A dietary supplement is a product (other than tobacco) that is intended to enhance the diet. It bears or contains one or a combination of dietary ingredients, such as a vitamin, mineral, herb, or other botanical product. Products have included approved new drugs, certified antibiotics, or licensed biologic agents that were marketed as dietary supplements or food before approval, verification, or licensure. These products are not recommended for use as a conventional meal or diet and are labeled as a dietary supplement. Subsequent to this bill’s passage, several synthetic AASs have become commercially available as dietary supplements; as such they do not have to pass the safety requirements of the FDA and are not held to strict quality control standards.⁶⁰ Claims regarding their effectiveness need not be substantiated by scientific proof as long as a disclaimer is listed on the product.

Dehydroepiandrosterone

DHEA is a naturally occurring hormone produced in the adrenal glands of the human body. Its synthetic form, isolated from soy and wild yams, is marketed as a supplement⁶¹ and has become popular for its antiaging properties, fat-burning properties, and enhancement of muscle mass, strength, and energy. DHEA is a weak androgen, but it is the most abundant steroid in the body serving as a precursor to testosterone, estrogen, progesterone, and corticosterone. Levels are high in the prenatal period as well as in puberty, gradually decreasing as an individual ages. Some studies performed on patients older than 50 years (when DHEA levels have dropped significantly) showed beneficial results after DHEA supplementation.⁶² Conversely, other studies have revealed inconsistent improvements in strength.⁶³

Many athletes use DHEA for its androgenic and anticatabolic effects. With its use come some undesired effects of feminization, as DHEA is a precursor to estrogen. Athletes attempt to combat this outcome by using clomiphene citrate (Clomid), an antiestrogenic drug, while taking DHEA. Little evidence exists to support claims for DHEA, either as a potent anabolic agent or as an antiaging drug.^{64,65} Furthermore, no studies of the long-term effects of taking DHEA have been published, particularly regarding the large doses used by athletes.⁶⁶

Androstenedione

Androstenedione, originally touted as the “secret weapon of the East Germans,” was first used in the 1970s, and its reputation as an effective anabolic agent encouraged worldwide use. In

March 2004, the US Department of Health and Human Services announced that the FDA had asked manufacturers to stop distribution of androstenedione. By October 2004, President Bush signed into law the Anabolic Steroid Control Act, which added androstenedione to the list of banned nonprescription steroid-based drugs. Currently Major League Baseball, the National Football League, the Olympics, and the National Collegiate Athletics Association (NCAA) prohibit its use. Androstenedione received tremendous exposure in the media during the 1998 baseball season when St. Louis Cardinals slugger Mark McGwire was discovered to be using the supplement, albeit legally at the time, after which sales of androstenedione rose dramatically.⁶⁷

Androstenedione is a potent AAS produced endogenously in the adrenal glands and gonads. In the liver, androstenedione is metabolized to testosterone. Similar to DHEA, during transformation it can be converted to testosterone and estrogen. As a result, a male athlete taking androstenedione can have elevated estrogen levels. One study observed that male subjects taking androstenedione experienced elevated estradiol levels equal to the estradiol levels seen in women during the follicular phase of the menstrual cycle, when levels are at their highest.⁶⁸ Additionally, as a testosterone precursor, it can lead to decreases in high-density lipoprotein cholesterol and enhance cardiac risk. Priapism has been described in an otherwise healthy young man using androstenedione who had no other precipitating factors; other concerns such as prostatic enlargement and cancer exist. Notably, significant ergogenic improvement after androstenedione supplementation has not been proved in the medical literature.

Doping

Historical Perspectives

The word *doping* originates from the Dutch word *dop*, an opium mixture used to enhance the racing capacity of horses. In 1967, the International Olympic Committee (IOC) published a medical code that included a list of banned drugs to “protect the health of athletes and to ensure respect for the ethical concepts implicit in Fair Play, the Olympic Spirit, and medical practice.”⁶⁹ The World Anti-Doping Agency (WADA) was established from this committee in 1999. Through its affiliation with various international, national, and private governing bodies, WADA has helped to establish strategies to tackle organized doping schemes, trafficking, and detection. It also publishes an annual listing of banned substances—an invaluable resource for athletes and medical providers.⁷⁰

Erythropoietin

WADA defines blood doping as “the misuse of certain techniques and/or substances to increase one’s red blood cell mass.”⁷⁰ Bonsdorff and Jalavisto⁷¹ were the first to discover and name the blood-stimulating hormone now commonly referred to as erythropoietin (EPO). Scientists later discovered its molecular structure, which served as a base for the development of recombinant human EPO (rhEPO). The first experiments with enhanced red blood cell mass and exercise showed that blood transfusion decreases submaximal heart rate for several weeks, predicting performance enhancement.⁷² Subsequent studies replicated the performance-enhancing effects of transfusion, including those

of Berglund and Ekblom who showed a 17% increase in the time to exhaustion among male athletes after 6 weeks of EPO administration.^{73,74} The IOC eventually banned blood transfusion for the 1988 Olympics.⁷⁵ In 1990, EPO was added to the IOC list of prohibited substances, and it has been on WADA's prohibited list since the organization's inception.

Mechanism of Action

Essential to athleticism and performance is how well trained or "conditioned" the athlete is. Basic physiology reminds us that muscle cells require oxygen to function optimally and that oxygen is transported throughout the body by erythrocytes. EPO is a hormone that stimulates the production of these erythrocytes in the range of 2.5×10^{11} erythrocytes per day, each of which has a life span of approximately 120 days.

Nearly 90% of EPO is produced in the peritubular renal cells, with the remainder being produced in the liver and brain. EPO production is triggered in response to hypoxia and stimulates an increase in erythrocyte production, leading to enhanced tissue oxygenation. Both epoetins and their synthetic analogues (erythropoiesis stimulating agents, or ESAs) have the ability to induce erythropoiesis by binding to EPO receptors on target cells.

Testing

WADA publishes an annual summary of prohibited substances and doping methods, the most recent having been released on January 1, 2017.⁷⁶ This list is commonly referred to as the Prohibited List, comprising a total of 10 different classes of banned substances (S0–S9), three different groups of prohibited methods (M1–M3), and two classes of drugs (P1 and P2) that are banned from selected sports only. The traceability of EPO abuse has been a complex issue, laborious and time-consuming, and frequently challenged by athletes. Drug testing in athletes is riddled with many hurdles, including cost. The inception of new ergogenic aids and novel masking agents has made the detection of illegal doping a challenge.⁷⁷ Allocation of innumerable research resources has been invested in improving analytical approaches as well as eliminating technical issues in detection. Given the negative ramifications of a positive drug test, it is critically important that the chain of custody be maintained and that testing protocols be updated and followed precisely to ensure the accuracy of sampling.

Detection of rhEPO includes both direct detection of recombinant isoforms and indirect approaches via the measurement of markers of enhanced erythropoiesis.⁷⁸ In the 1990s, the International Cycling Union introduced random blood tests to detect abnormalities in biologic parameters as a screening test for subsequent urinary detection of rhEPO. Although this technique was successful in preventing heavy use of rhEPO, it proved unable to discriminate against athletes with naturally elevated blood parameters. Individual reference ranges were introduced in 2007 in the form of "biologic transport," or the method of tracking each athlete's own blood numbers, known as the Athlete Biologic Passport hematologic evaluation. Markers that are currently tracked include hematocrit, hemoglobin, red blood cell count, reticulocyte percentage, reticulocyte number, mean corpuscular volume, mean corpuscular hemoglobin, and mean corpuscular

hemoglobin concentration, among others. Blood results are then processed by a model that identifies any abnormal blood parameters as compared with the athlete's individual baseline. These operating guidelines, in accordance with the WADA code, were implemented in 2009.

Newer generations of these models have since been employed to include advances in electrophoresis techniques—isoelectric separation, Western blot analysis, antibody detection, and mass spectrometry—to help distinguish between physiologic EPO versus rhEPO. Although there is still a lack of total understanding, research continues in an attempt to further develop new testing methods, decrease false positives, and improve the sensitivity of the current testing measures. What is clear is that the combination of an athlete's biologic passport (based on individual profiling) and urine tests targeting rhEPO has undoubtedly made it more challenging for athletes to compete illegally.⁷⁹

Side Effects

To some athletes, the benefits of using performance-enhancing agents clearly outweigh any risk. EPO and its synthetic analogues are known to increase red blood cell mass. Physiologic benefits include normalization of cardiac output and enhanced immune function, leading to improved exercise tolerance, reduced fatigue, and subjective improved quality of life. EPO excess has been associated with hazardous adverse effects, including increased blood viscosity. This hyperviscosity, in turn, can lead to headache, hypertension, congestive heart failure, venous thromboses, pulmonary emboli, encephalopathy, and stroke. There have been some reports that rhEPO administration leading to excessive erythropoiesis may result in reduced thickness of cortical bone, trending toward osteoporosis and theoretically increased fracture risk.⁸⁰

Creatine

Historical Perspectives

Creatine is one of the most popular supplements currently used by athletes. It is a compound synthesized within the body and also absorbed exogenously from fish and meats. In 1992, creatine was manufactured as a supplement to potentially improve muscular performance. Shortly thereafter, it was touted as a supplement that delayed fatigue and enhanced athletic performance.⁸¹

Physiology

Creatine has a fundamental role in the structure and function of adenosine triphosphate, the body's prime energy source. It is synthesized largely in the liver from amino acids and subsequently transported to skeletal muscles, the heart, and the brain, where it is absorbed. Intracellularly, creatine is phosphorylated into creatine phosphate, and in this state it serves as an energy substrate that contributes to the resynthesis of adenosine triphosphate for energy.⁸² Creatine is also proposed to serve as a buffer to lactic acid produced during exercise. Lactic acid is a byproduct of intense exercise caused by the accumulation of H⁺ molecules. In excess, it limits exercise. Creatine binds these molecules, delays fatigue, and lengthens exercise duration. Creatine may also help athletic performance by promoting protein synthesis, thereby increasing muscle mass.⁸¹ Additionally, when combined with a

high-carbohydrate diet, creatine enhances glycogen stores, which is advantageous during long-duration, high-intensity exercise.⁸³

Creatine may benefit athletes who engage in sports that emphasize short bursts of intense anaerobic activity, such as football, soccer, lacrosse, hockey, basketball, and powerlifting. Short-term creatine supplementation has been shown to enhance the ability to maintain muscular forces during jumping,⁸⁴ intense cycling,⁸⁵ and weight lifting.⁸¹ It is important to note, however, that not all athletes appear to benefit from creatine supplementation.^{83,86} Some individuals are “responders” while others are “nonresponders” to creatine.⁸⁷ It is estimated that about 30% of athletes are “nonresponders” because they already naturally possess a maximal amount of creatine—as much as can be stored by their muscle cells. Muscle creatine content is measured only by muscle biopsy, making it necessary for an athlete to use the supplement on a “trial and error” basis to determine if it will be beneficial.

Early studies emphasized the importance of a loading phase with creatine. Athletes would use large doses (20 g/day of creatine for 5 days) before reaching a maintenance dose (2 g/day).⁸⁸ More recent evidence has shown that loading is unnecessary and that a low-dose regimen of 3 g/day is just as effective, although it may take longer for benefits to be noted.^{89,90}

Adverse Effects

An increase in body mass of up to 2 kg occurs frequently and is believed to be the result of water retention.^{83,91} According to some recommendations, creatine should not be consumed either before or during intense exercise.⁹² Case reports of gastrointestinal distress (diarrhea) are common, but a direct cause-and-effect relationship has not been defined. Concerns regarding nephrotoxicity have been raised, but creatine supplementation does not appear to negatively affect renal function in healthy persons without a previous history of renal disorder if they adhere to recommended dosing.⁹³

Thus the use of creatine in athletes is popular because it is considered a legal means of improving performance. Despite findings from a few controversial studies, supplementation with creatine can increase fat-free mass and strength. Creatine may also be of benefit to athletes who engage in high-intensity sprints or endurance training, but these benefits seem to diminish with prolonged exercise.⁸³

Growth Hormone

Historical Perspectives

One of the most popular ergogenic drugs used by athletes, human GH, was discovered in the 1920s, when researchers noted that after an injection of ox pituitary glands, normal rats grew abnormally large. Animal breeders later used these injections to increase muscle mass and decrease body fat in their breeds.⁹⁴ In the 1950s, GH extracted from the brains of cadavers from Africa and Asia was administered to children whose growth was stunted by absence of this hormone.⁹⁵ Thousands of short-statured children were successfully treated with human GH. However, as a result of treatment with cadaver extracts, Creutzfeldt-Jakob disease developed in many of these children. This disease causes progressive dementia, loss of muscle control, and death.⁹⁶ By the early 1990s,

the FDA had stopped distribution of GH, yet there was a need to treat children who had short stature induced by low levels of GH. The Genentech Company used recombinant deoxyribonucleic acid technology to manufacture a biosynthetic GH, somatrem (Protropin), enabling GH to be safely provided to children in need. However, the use of human GH as a supplement is universally illegal.

GH affects almost every cell in the body. Because human GH levels dip with advancing age, it has been endorsed as an antiaging product. Human GH secretion can be stimulated throughout life by either sleep or exercise; thus combining exercise with rest may be beneficial. Currently long-term studies on the effectiveness of human GH are lacking.

Mechanism of Action

Human GH is a polypeptide produced and stored in the anterior pituitary gland. Its secretion is high during puberty and it continues to be secreted throughout life. Secretion occurs daily in a pulsatile fashion, with the highest levels observed shortly after sleep initiation.⁹⁴ Human GH works in two ways: first, by binding on target cells, it exerts its effects on adipocytes, which, in turn, break down triglycerides and prevent lipid accumulation; second, after reaching the liver, human GH is rapidly converted into insulin-like growth factor-1 and thus exerts its growth-promoting effects throughout the body.

Athletes have used human GH and insulin-like growth factor-1 as ergogenic supplements. Human GH is an appealing supplement for athletes because it stimulates protein metabolism. It is also a potent regulator of carbohydrate metabolism by decreasing insulin sensitivity and cellular uptake of glucose. In addition, human GH stimulates the catabolism of lipids, making free fatty acids available for quick energy use and sparing muscle glycogen.⁹⁷ Human GH also stimulates bone growth, which is crucial for growth in prepubertal youths.

Adverse Effects

Most notably, excess GH may cause problems with bone growth in adults. In adults, growth plates have fused but continue to enlarge in the presence of high levels of GH, resulting in dysmorphic, acromegalic features, particularly of the face, hands, and feet. Other problems include a predisposition to diabetes (attributed to decreased insulin sensitivity), cardiomyopathy, and congestive heart failure.⁹⁷ Correlations have been noted between human GH and the enlargement of intracranial lesions, amplification of intracranial hypertension, and leukemia in patients treated with recombinant human GH.⁹⁸

No scientific studies have been conducted to show improved athletic performance with use of human GH. Nevertheless, human GH has been widely used by athletes. Limitations of human GH include poor quality (because much of it is supplied from international sources), exorbitant cost (about \$1000 per month), and availability only in the parenteral form (increasing risk of infection). Nonetheless, human GH is used frequently, especially because few solid testing methods exist aside from two tests performed within a few days of dosing. The isoform test detects the presence of synthetic human GH and is effective for detection of use within 12 to 72 hours of administration. The second

test, called the “biomarker test,” evaluates the presence of chemicals produced by the body after human GH use⁹⁹; it may be used alone or in conjunction with the isoform test.

Caffeine

Historical Perspectives

Caffeine is the most widely used legal ergogenic drug, with billions of people using it. The United States Olympic Committee (USOC) had previously placed caffeine in a restricted status, allowing no more than 12 µg/mL in urine; however, because of highly variable excretion rates, this restriction has been eliminated. Doses between 5 and 10 mg/kg 1 hour before an athletic contest are commonly used. Caffeine is not included on the 2017 Prohibited List of substances, having been removed since 2004. The drug is part of WADA’s monitoring program, a program that includes substances not prohibited from sport but monitored to detect misuse in sport. Since 2010, caffeine use among athletes has increased, although there have been no global indications of misuse.⁷⁵

Mechanism of Action

Caffeine has historically been used to enhance physical and mental performance. It employs its ergogenic effects both centrally and peripherally. Once in the bloodstream, caffeine is rapidly absorbed, leading to the stimulation of excitatory neurotransmitters. At a central location, caffeine increases perception of physical effort and improves neural activation of muscle transport. Peripherally, caffeine leads to the release of free fatty acids from adipose tissue, sparing muscle glycogen and maintaining blood glucose levels. It also stimulates potassium transport into tissue, maintaining muscle cell membrane excitability and decreasing muscle cell reaction time after neural stimulus, while reducing muscle fatigue.

Side Effects

Evidence indicates that caffeine ingestion helps with endurance while providing performance enhancement.¹⁰⁰ Beneficial effects occur at modest doses of about 1 to 3 mg/kg 1 hour prior to exercise without noticeable dose dependency, although higher doses (>6 to 9 mg/kg) tend to incur adverse side effects. The most common detrimental effects include anxiety, restlessness, panic attacks, gastritis, reflux, and heart palpitations.¹⁰¹ Chronic users may experience withdrawal, including headache and fatigue with abrupt cessation of use. Acute caffeine intake is known to increase urinary loss of fluid, although a recent review proposes that there is little evidence that the drug affects overall fluid status.¹⁰² Doses of caffeine within the ergogenic range do not alter sweat rates, urinary losses, or indices of hydration status.¹⁰³

The majority of studies highlighting caffeine use and athletic performance concern endurance sports. In long-distance cycling competitions, time to exhaustion at VO₂ max was significantly improved.¹⁰⁴ In another cycling study, a 7% increase in distance covered in 2 hours was noted.¹⁰⁵ There is also evidence that caffeine can enhance performance in sustained, high-intensity activities. In one particular study, 1500-meter swim time was significantly improved in trained athletes.¹⁰⁶ There is a strong need to educate athletes regarding the responsible intake of caffeine

to appropriately balance and maximize benefit without eliciting the negative side effects of use.

Summary

For centuries, athletes have been willing to risk death in an attempt to improve their ability to compete in sports. What can persons entrusted with the health of athletes do to protect athletes from themselves? The first step is “meaningful education” as reflected by an important tenet in medicine—“you only recognize what you know.” Most medical personnel receive very little education about doping and ergogenic aids during their schooling. To stay abreast with the practices of athletes, lifelong learning is essential. Health care professionals should be able to communicate openly with athletes regarding the health risks and legality of ergogenic aids and the importance of healthy nutrition and honest training as a viable and preferred alternative for promoting athletic superiority. Second, the medical community should conduct scientific controlled studies on athletes who are using ergogenic drugs. Obviously such studies would be difficult to conduct because many of these drugs are illegal, many ergogenic agents have significant adverse effects, and research oversight committees are often reluctant to approve studies using drugs with known or suspected adverse effects. Third, effective drug testing and stringent penalties for athletes, coaches, teams, and nations who use banned substances are imperative.

SPORTS PHARMACOLOGY: RECREATIONAL DRUG USE

Substance use among students and young adults continues to be a leading cause of morbidity and mortality. The Monitoring the Future Survey is an ongoing study of the behaviors, attitudes, and values of American secondary school students, college students, and young adults.¹⁰⁷ The 2016 updated survey shows that nearly 50% of young Americans have used some type of illicit substance between grades 8 and 12, with nearly 25% having used such a substance in the past 12 months. Cigarette smoking and alcohol use have continued to decline and are at all-time lows. Even so, vaping (the inhalation of vapors such as e-cigarettes), alcohol, and marijuana remain the most consistently abused substances. Consistent with prior studies, there appears to be little difference in the prevalence of recreational drug use in athletes and nonathletes. Clinicians need to be aware of the effects of these drugs in order to appropriately educate their athletes on the risk of these substances not only on performance measures but, more importantly, also on general health and wellness.

Alcohol

Epidemiology

Alcohol continues to be one of the most frequently abused substances in collegiate, professional, and Olympic sports.¹⁰⁸ The use of alcohol has declined since 1980, yet despite this fact, the National Institute of Drug Abuse reports that in 2016, approximately 33% of high school seniors acknowledged having consumed alcohol in the previous month.¹⁰⁹ According to the Harvard School of Public Health Alcohol Study, 80% of college

students consume alcohol. Compared with nonathletes, collegiate athletes exhibit a higher rate of binge drinking; they also experience more alcohol-related complications including academic problems, driving under the influence, gambling, illicit drug use, and sexual promiscuity.^{110,111}

Pathophysiology and Adverse Effects

Ethanol is the primary psychoactive constituent in alcoholic beverages; it exerts its effect on the central nervous system (CNS) via its action on gamma-aminobutyric acid receptors. It also interacts with acetylcholine, serotonin, and N-methyl-D-aspartate receptors. It is oxidized by the liver at a constant rate rather than having an elimination half-life. Systemic ethanol levels are typically quantified by the blood alcohol content (BAC). Adverse effects of ingestion often correlate with BAC levels; however, persons lacking effective forms of metabolizing enzymes may experience more severe symptoms, whereas those who have acquired tolerance may metabolize alcohol more rapidly. In general, persons with a low BAC level (0.05%) experience euphoria, talkativeness, and relaxation. BAC levels of 0.1% or higher frequently induce CNS depression, compromise motor and sensory function, and impair cognition. Levels in excess of 0.3% may lead to unconsciousness, with levels surpassing 0.4% placing individuals at risk of death.

In 1982, the American College of Sports Medicine issued a position statement regarding alcohol and its effect on sports performance.¹¹² The main points of the position statement are as follows:

1. Alcohol adversely affects coordination, balance, and accuracy.
2. Alcohol does not improve athletic performance.
3. Alcohol will not improve muscle work performance and negatively affects the ability to perform.
4. Alcohol may impair temperature regulation during prolonged exercise in a cold environment.

Other Considerations

In recognition of the epidemic of alcohol abuse among young Americans, the Surgeon General's Office released its Call to Action against Underage Drinking in 2007. This document identified the following six goals to help lower the incidence of alcohol abuse and associated morbidity and mortality.¹¹³

1. Foster changes in society that facilitate healthy adolescent development and that help prevent and reduce underage drinking.
2. Engage parents, schools, communities, all levels of government, all social systems that interface with youths, and youths themselves in a coordinated national effort to prevent and reduce underage drinking and its consequences.
3. Promote an understanding of underage alcohol consumption in the context of human development and maturation that takes into account individual adolescent characteristics, along with environmental, ethnic, cultural, and gender differences.
4. Conduct additional research on adolescent alcohol use and its relationship to development.
5. Work to improve public health surveillance on underage drinking and on population-based risk factors for this behavior.

6. Work to ensure that policies at all levels are consistent with the national goal of preventing and reducing underage alcohol consumption.

Despite campaigns to decrease binge drinking by athletes, excessive drinking is a reality. Team physicians should work closely with coaching staffs and athletic departments to develop prevention programs and an action plan for penalties for alcohol-related infractions of established rules.

Marijuana

Epidemiology

Marijuana is the second most commonly abused drug by athletes and the most commonly used illicit drug, with nearly 1 in 15 high school seniors reporting daily or nearly daily use.^{114,115} According to the most recent National Survey on Drug Use and Health released by the CDC in 2015, an estimated 22.2 million Americans aged 12 years or older are current users. Though this is not an increase, use in individuals age 18 or older is on the rise. Despite legalization across several states, the NCAA continues its stance in considering this a banned substance. Nevertheless, the use of marijuana is still pervasive and persisting among student athletes.

Pathophysiology and Adverse Effects

All forms of cannabis are mind-altering (psychoactive) drugs. The main active chemical derivative, Δ -9-tetrahydrocannabinol, affects cannabinoid receptors found in the brain and peripheral tissues and also indirectly enhances dopamine release, producing its psychotropic influences. Marijuana is absorbed readily and has effects throughout the body, most predominantly in the CNS, the respiratory system, and the cardiovascular system.

Short-term effects of marijuana use include distorted perception, difficulty in thinking and problem solving, and loss of coordination. The psychotropic effects of cannabis include general euphoria and a mild release from inhibitions, drowsiness, stimulated appetite, and freedom from anxiety. Other persons experience an altered sense of consciousness.¹¹⁶ Long-term use may be linked to behavioral changes similar to those caused by traumatic injury.

Recent studies document a spectrum of cardiorespiratory ailments associated with the smoking of marijuana. A large epidemiologic study suggests that marijuana smoke can cause the same types of respiratory damage as tobacco smoke.¹¹⁷ The acute physiologic effects of marijuana on the cardiovascular system include a substantial dose-dependent increase in heart rate, a mild increase in blood pressure, and occasionally orthostatic hypotension. Athletic performance is affected, as marijuana use has been shown to decrease exercise test duration in maximal exercise tests, with premature achievement of maximal oxygen uptake.¹¹⁸

Other Considerations

Discussions with athletes should not only emphasize the negative effects of marijuana on athletic performance but also stress the often forgotten fact that marijuana continues to be a Schedule I drug despite legalization in many states. Moreover, more potent and dangerous synthetic derivatives have been developed and are

increasingly popular in attempts to bypass screening tests. These products, such as “Spice” or “K-2,” are herbal mixtures infused with chemicals to give effects similar to those of marijuana by their ability to bind to cannabinoid receptors throughout the body. Their popularity stems from the “marijuana-like high” and easy availability at retail outlets and via the Internet. Numerous reports of serious adverse effects—including convulsions, anxiety attacks, tachycardia, vomiting, psychosis, and disorientation—have necessitated their emergency classification in March 2011 as a Schedule I substance.¹¹⁹

Tobacco

Epidemiology

Smoking remains the most preventable cause of death in developed countries. The World Health Organization reports that tobacco use is responsible for more than 7 million deaths annually worldwide.¹²⁰ The use of tobacco by adolescents has declined since the peak levels of the mid-1990s; however, 15% of individuals in the United States continue to smoke. In addition to cigarette smoking, other forms of tobacco use include smokeless tobacco, pipes, cigars, chewing tobacco, snuff, and, most recently, the e-cigarette. In 2014, sales of e-cigarettes surpassed those of traditional cigarettes. In a recent press release regarding the e-cigarette epidemic, the Surgeon General reported that 1 out of every 6 high school students in the United States has used e-cigarettes in the previous month.¹²¹ Although cigarette smoking is less of a problem in athletes than in nonathletes, use of smokeless or “spit” tobacco remains prevalent, particularly in the realms of baseball, football, and golf.

Pathophysiology and Adverse Effects

Nicotine is both a stimulant and a depressant. As nicotine enters the body, it crosses the blood-brain barrier within 10 to 20 seconds and exerts its effects by binding to nicotinic acetylcholine receptors, increasing levels of dopamine. Via this mechanism, tobacco is believed to have an addictive potential comparable to that of alcohol, cocaine, and morphine.¹²²

The detrimental health consequences of tobacco use are well known. Smokeless tobacco users have an increased risk for oral cancer, including cancer of the lip, tongue, cheeks, gums, and the floor and roof of the mouth. Through its release of various chemical messengers—including acetylcholine, epinephrine, norepinephrine, serotonin, and dopamine, among others—nicotine appears to enhance concentration, memory, and alertness while decreasing appetite and promoting relaxation. Nicotine is considered a mood- and behavior-altering drug, often helping athletes to deal with stressful situations. Other effects, due to nicotine’s vasoactive properties, include increases in pulse rate by 10 to 20 beats/min and blood pressure by 5 to 10 mm Hg. Nicotine also stimulates platelet aggregation, which may increase the risk for blood clots.

Other Considerations

In an effort to curtail tobacco use, both professional baseball (minor leagues) and junior hockey (Western Hockey League) have banned the use of spit tobacco by players, coaches, and officials. Similarly, the NCAA bans the use of spit tobacco by

players, coaches, and officials during NCAA-sanctioned events. Dental screening, which can often detect precancerous lesions that may be caused by nicotine, is highly recommended for all athletes. Last, nicotine addiction is difficult to overcome. Epidemiologic studies indicate that the likelihood of addiction increases when cigarette smoking commences during adolescence. Thus addressing the dangers of nicotine in youth sports and its potential for refractory addiction is critical for prevention.

Cocaine

Epidemiology

The Monitoring the Future Survey documented that in 2006, 8.5% of 12th graders had used cocaine.¹²³ Since then, use has dropped to a historical low of 3.7% among this same cohort.¹²³ Cocaine abuse can lead to physical dependence. It is a controlled substance (Schedule II) and is banned in sport, with the IOC testing athletes routinely.

Pathophysiology and Adverse Effects

Cocaine is a naturally occurring alkaloid that is present in the leaves of *Erythroxylon coca*. It is commercially available and can be applied to mucous membranes of the oral, laryngeal, and nasal cavities for use as a topical anesthetic; however, it is most known for its abuse potential via its various processed forms. Cocaine exerts its adverse effects as a result of excessive sympathetic activity. Unlike other local anesthetics, cocaine affects the nervous system by potentiating catecholamines and leads to increased energy levels. It also results in the inhibition of pre-synaptic reuptake of norepinephrine, dopamine, and serotonin.

Cocaine causes an acute dopamine release and inhibits synaptic dopamine reuptake, which produces euphoria, reduced fatigue, elevated sexual desire, and increased mental ability as well as sociability. At higher doses, tremors and tonic-clonic convulsions may occur. Long-term use of cocaine can result in nasal congestion, rhinitis, chronic sinusitis, and increased risk for upper respiratory infection. CNS toxicity is extremely common with cocaine use, including agitation, anxiety, apprehension, confusion, headache, dizziness, emotional lability, euphoria, excitement, hallucinations, seizures, and psychosis.

Effects on Athletic Performance

The significant morbidity and mortality in athletes is a result of the effects of cocaine on the cardiovascular system. Through its sympathomimetic effects, cocaine augments ventricular contractility, blood pressure, heart rate, and myocardial oxygen demand. Myocardial ischemia is induced by coronary vasoconstriction, platelet aggregation, and accelerated atherosclerosis. The ensuing supply-demand deficit may manifest as angina, and cocaine-induced coronary spasm may lead to acute myocardial infarction and death. Cocaine is markedly pyrogenic because it induces muscular activity, augments heat production, and potentiates vasoconstriction—all of which can enhance the risk for heat stroke and death. Cocaine dependence is a significant problem in our society; it is associated not only with the outlined spectrum of medical complications but also with crime and violence. Educational programs regarding the hazards of cocaine use and addiction are highly recommended for all athletes.

Inhalants

Epidemiology and Classification

Inhalant use is the deliberate inhalation of volatile substances to induce a mind-altering effect. Purchase and possession are legal, and inhalants are cheap and easily accessible. National surveys of adolescents in the United States have reported that second to marijuana, inhalants are the most widely used class of illicit drugs among 8th and 10th graders. Fortunately, their use has continued to decrease. The American Academy of Pediatrics has taken the initiative to educate clinicians, parents, and children of the dangers of inhalant abuse.¹²⁴ Inhalants are divided into three groups on the basis of the inhalant pharmacology, as follows:

1. Type I agents include volatile solvents such as paint thinner, acetone, glue, rubber cement, butane, aerosols, hair spray, and gasoline.
2. Type II agents are nitrous oxide and whipping cream aerosols, which contain nitrous oxide.
3. Type III agents include volatile alkyl nitrites known as “poppers,” “snappers,” “boppers,” and “Amys.”

Inhalant agents are abused through a variety of methods. Glue or solids are emptied into a bag, held close to the nose, and inhaled either through the nose (*snorting or sniffing*) or through the mouth (*huffing*). Other methods include placing one's entire head into a bag (*bagging*) and inhalation of either air fresheners (e.g., Glade, referred to as *glading*), or computer cleaning aerosols (*dusting*).

Pathophysiology and Adverse Effects

Inhalants contain different solvents, each with its own unique toxicity, but all inhalants induce CNS depression, likely involving a γ -aminobutyric acid agonist or altered neuronal membrane function.¹²⁵ Death can occur as a result of the use of inhalant agents because of cardiac complications that occur as a result of asphyxia, ventricular fibrillation, or cardiac arrhythmia. Cerebral death may occur as a result of asphyxia, edema, and hyperpyrexia.¹²⁵

Inhalants do not augment or improve athletic performance; in fact, their use can lead to serious health consequences. Some glues are metabolized and may cause peripheral neuropathy, characterized by muscle weakness and wasting. Inhalants have also been known to sensitize and potentiate the myocardium to epinephrine, which, during exercise, may increase an athlete's risk of an acute, fatal arrhythmia. No specific antidotes exist for inhalant toxicity, and treatment is rooted in prevention.

Conclusion

The use of recreational drugs and their potential for deadly consequences are a realities for athletes and our society. Most treatment entails meaningful educational programs. Athletes who are well informed about the potential dangers of using recreational drugs may be best able to avoid the temptation to “try” these drugs, which are readily available. Additionally, coaches should be involved in the education process; they should stress

the importance of sobriety and healthy habits as core requirements for optimal health and the development of a successful team. The ongoing Monitoring the Future surveys have clearly demonstrated that all the drugs discussed in this chapter are used by many youth at very young ages, often in middle or elementary school. It is therefore imperative that the education of athletes, coaches, and parents commence in the earliest recreational leagues.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Cooper R, Naclerio F, Allgrove J, et al. Creatine supplementation with specific view to exercise/sports performance: an update. *J Int Soc Sports Nutr.* 2012;9:33.

Level of Evidence:

II

Summary:

More and more information is being discovered about the use of creatine, particularly with regard to its relevance in improving sports performance. This article reviews recent advances in understanding the science relating to creatine use and the methods of supplementation to make cellular and subcellular changes.

Citation:

Green GA, Uryasz FD, Petr TA, et al. NCAA study of substance use and abuse habits of college student-athletes: clinical investigations. *Clin J Sport Med.* 2001;11(1):51–56.

Level of Evidence:

III

Summary:

Monitoring the Future is an ongoing study that entails the administration of a series of annual surveys among representative samples of secondary school students throughout the United States, using a standard set of questions to determine usage levels of various substances. Cigarettes/e-cigarettes, alcohol, and marijuana continue to be the most frequently abused substances among adolescents, with synthetic marijuana recently falling under the Drug Enforcement Administration's list of prohibited substances.

Citation:

World Anti-Doping Agency. The 2012 Prohibited List. Available at: http://www.wada-ama.org/Documents/World_Anti-Doping_Program/WADP-Prohibited-list/2012/WADA_Prohibited_List_2017. Accessed June 12, 2017.

Level of Evidence:

V

Summary:

The World Anti-Doping Agency annually updates its list of prohibited substances among athletes. A review of and familiarity with the list is critical for providers so they can comply with up-to-date guidelines.

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